AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB)

Faculty of Science and Technology (FST)
Department of Computer Science (CS)
Undergraduate Program

COURSE PLAN	**SEMESTER: Spring 2025-20256**
I. Course Code and Title CSC 2210 Object Oriented Programming 2 **II. Credit** 3 Credits (2hrs theory and 2hrs 20minutes Lab per week) **III. Nature** Core Course for CSE and COE **IV. Prerequisite** CSC 2209: Object Oriented Programming 1	**V. Vision:** Our vision is to be the preeminent Department of Computer Science through creating recognized professionals who will provide innovative solutions by leveraging contemporary research methods and development techniques of computing that is in line with the national and global context. **VI. Mission:** The mission of the Department of Computer Science of AIUB is to educate students in a student-centric dynamic learning environment; to provide advanced facilities for conducting innovative research and development to meet the challenges of the modern era of computing, and to motivate them towards a life-long learning process.

VII - Course Description

- Contrast the Fundamentals of Object-Oriented Programming and learn .Net Framework 4.7.2.
- Create .Net Technology applications that leverage the object-oriented features of the C# language, such as encapsulation, inheritance, polymorphism, and abstraction.
- Formulate and derive .Net technology data types and expressions.
- Relate the use of arrays and other data collections for real life application.
- Explain the concept of Assembly and present the working principle of the compiler.
- Build and justify error-handling techniques using exception handling.
- Compose and manipulate multiple operations on database tables, including creating, reading, updating, and deleting using MSSQL.
- Create an event-driven graphical user interface using Windows Form: panels, buttons, labels, text fields, and text areas.

VIII - Course outcomes (CO) Matrix

By the end of this course, students should be able to:

COs*	CO Description	Level of Domain***			PO Assessed ****
		C	P	A	
CO1 **	**Perform** the proper way of presenting and writing the Object-Oriented Programming according to the major Principles in analytical problem solving.			2	PO-j-1
CO2 **	**Display** and **verify** the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.			5	PO-k-3
CO3 **	**Prepare** and **Explain** a real life desktop based application **synthesizing** several component of C# along with development tools to adhere the given requirements.			4	PO-j-3

CO4	**Perform** and **solve** the proper system functional requirements which combines several aspects of the work to display and verify a real-life based application.			5	PO-k-3

C: Cognitive; P: Psychomotor; A: Affective Domain

* *CO assessment method and rubric of COs assessment is provided in later section*

** *COs will be mapped with the Program Outcomes (POs) for PO attainment*

*** *The numbers under the 'Level of Domain' columns represent the level of Bloom's Taxonomy each CO corresponds to.*

**** *The numbers under 'PO Assessed' column represent the POs each CO corresponds to.*

IX - Topics to be covered in the class and/or lab: *

Time Frame	CO Mapped	Topics	Teaching Activities	Assessment Strategy(s)
Week 1	CO1	Mission & Vision of AIUB and Faculty of Science & Technology, Introduction to Visual Studio .Net Environment, C# Basics, Discussion about OBE and it's usefulness.	Lecture, Lab Work	Term Exam
Week 2	CO1, CO3	Params, ref, out, Class, Object	Lecture, Lab Work	Term Exam
Week 3	CO1, CO3	Constructor, Properties, OOP principles	Lecture, Lab Work	Term Exam
Week 4	CO1, CO3	Encapsulation	Lecture, Lab Work, Quiz 1	Term Exam
Week 5	CO3	Array	Lecture, Lab Work	Term Exam
Week 6	CO1, CO3	Inheritance, Polymorphism	Lecture, Lab Work	Term Exam
Midterm (Week 7)				
Week 8	CO2, CO3	Abstraction and other types	Lecture, Lab Work	Project
Week 9	CO2, CO3	Association, Interface	Lecture, Lab Work	Project
Week 10	CO2, CO3	Generics, List, Indexer, Events and Delegates	Lecture, Lab Work	Project
Week 11	CO2, CO3, CO4	Database concepts, Windows Form	Lecture, Lab Work, Quiz 1	Project
Week 12	CO2, CO3, CO4	Project Discussion	Project	Project
Week 13	CO2, CO3, CO4	Project Submission and Viva	Project	Project
Final term (Week 14)				

* *The faculty reserves the right to change, amend, add, or delete any of the contents.*

X - Mapping of PO to Courses and W, K, P, A

POI	PO Indicators Definition	Domain	W	K	P	A
PO-j-1	Optimize computer science and engineering solutions by giving and responding to clear instructions. (Communicate effectively by giving and responding to clear instructions to produce computer science and engineering solutions.)	Affective Level 2 (Responding)	0.4			A1 A3 A5
PO-k-3	Demonstrate competency in completing individual computer science and engineering project based on relevant management principles and economic models.	Affective Level 5 (Characterization)	0.4			
PO-j-3	Make and deliver effective presentation based on complex computer science and engineering activities.	Affective Level 4 (Organizing)	0.3			A1 A2

XI – K, P, A Definitions

Indicator	Title	Description
A1	Range of resources	Involve the use of diverse resources (and for this purpose resources include people, money, equipment, materials, information, and technologies)
A2	Level of interaction	Require resolution of significant problems arising from interactions between wide-ranging or conflicting technical, engineering, or other issues
A3	Innovation	Involve creative use of engineering principles and research-based knowledge in novel ways
A5	Familiarity	Can extend beyond previous experiences by applying principles-based approaches

XII – Mapping of CO Assessment Method and Rubric

The mapping between Course Outcome(s) (COs) and The Selected Assessment method(s) and the mapping between Assessment method(s) and Evaluation Rubric(s) is shown below:

COs	Description	Mapped POs	Assessment Method	Assessment Rubric
CO1	Perform the proper way of presenting and writing the Object-Oriented Programming according to the major Principles in analytical problem solving.	PO-j-1	Term Exam, Lab Work	Rubric for Term Exam, Lab work
CO2	Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.	PO-k-3	Project, Report	Rubric for Project, Report
CO3	Prepare and Explain a real life desktop based application synthesizing several component of C# along with development tools to adhere the given requirements.	PO-j-3	Project, Report	Rubric for Project, Report
CO4	Perform and solve the proper system functional requirements which combines several aspects of the work to display and verify a real-life based application. Besides, Defend the system based on market demand by delivering an effective presentation to the external	PO-k-3	Project, Viva (QA)	Rubric for Project, Viva (QA)

	and audience.			

XIII – Evaluation and Assessment Criteria

CO1: Perform the proper way of presenting and writing the Object-Oriented Programming according to the major Principles in analytical problem solving.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Problem Analysis	Fails to provide a description of the problem or inaccurately identifies the real problem. There is a lack of clarity regarding the relevance of OOP principles in solving the problem, and their necessity is not addressed.	Offers a superficial description of the problem with limited clarity. The real problem is vaguely identified, and the explanation of why examining it is crucial lacks depth. The need for OOP principles in solving the problem is mentioned but not thoroughly explained.	Provides a basic description of the problem, but it lacks full clarity or detail. The explanation of the real problem's significance is somewhat clear but could be more comprehensive. The relevance of OOP principles to solve the problem is adequately explained but lacks depth.	Offers a clear and concise description of the problem, highlighting its significance within the context of Object-Oriented Programming Principles. The explanation of why examining the problem is crucial is articulate, and the necessity of OOP principles in solving it is well-justified.	Provides a thorough and insightful description of the problem, clearly demonstrating its relevance to OOP principles. The explanation of why examining the problem is crucial is nuanced and comprehensive , and the necessity of OOP principles in solving it is eloquently justified with sophisticated reasoning.
Content Knowledge	Does not demonstrate an understanding of Object-Oriented Programming (OOP) principles or their advantages in problem-solving. There is a lack of relevant content knowledge, and the response shows significant inaccuracies or misunderstandings regarding OOP concepts and their application.	Demonstrates limited understanding of Object-Oriented Programming principles and their advantages in problem-solving. The response lacks depth and coherence, and there are notable gaps or inconsistencie s in the explanation of	Demonstrates a basic understanding of Object-Oriented Programming principles and their advantages in problem-solving. The response provides a somewhat accurate explanation of OOP concepts and their application but may lack detail or fail to	Demonstrates a solid understanding of Object-Oriented Programming principles and their advantages in problem-solving. The response offers a clear and well-structured explanation of OOP concepts, supported by relevant examples, and effectively highlights the	Demonstrates an exceptional understanding of Object-Oriented Programming principles and their advantages in problem-solving. The response is insightful, and demonstrates a sophisticated grasp of OOP concepts, with articulate explanations and nuanced

		OOP concepts or their application to problem-solving scenarios.	address all aspects of the question comprehensively.	benefits of using these principles in solving problems.	insights into the benefits of applying these principles to real-world problems.
Completeness	Fails to demonstrate any understanding of Object-Oriented Programming principles. There is no attempt to solve the problem criteria, and the necessary steps are completely omitted or incorrectly applied.	Demonstrates limited understanding of some Object-Oriented Programming principles. They attempt to solve some of the problem criteria but fail to address all aspects adequately. Some necessary steps may be missing or improperly executed.	Shows a basic understanding of most Object-Oriented Programming principles. They attempt to solve most of the problem criteria, but some aspects lack depth or clarity. Most necessary steps are completed, but there may be minor omissions or errors.	Demonstrates a solid understanding of Object-Oriented Programming principles. They effectively solve the problem criteria, addressing all aspects comprehensively and accurately. All necessary steps are completed with clarity and coherence.	Exhibits an exceptional understanding of Object-Oriented Programming principles. They expertly solve the problem criteria, providing insightful and comprehensive solutions. All necessary steps are completed with precision, demonstrating a mastery of Object-Oriented Programming principles.

CO2: Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)
Evaluation Criteria	**Evaluation Definition**				
Requirement fulfillment	Fails to demonstrate any understanding of real-life scenario-based project development or functional requirement identification. There is no attempt to depict a project or identify functional requirements accurately.	Demonstrates limited understanding of real-life scenario-based project development and functional requirement identification. The project depicted lacks coherence or relevance to real-life scenarios, and functional requirements are inaccurately identified or	Presents a basic depiction of a real-life scenario-based project and identifies some functional requirements. However, the project lacks depth or complexity, and some functional requirements may be vaguely defined or missing key details.	Effectively demonstrates a realistic scenario-based project and accurately identifies most functional requirements. The project is well-developed with appropriate complexity, and functional requirements are clearly articulated with relevant details.	Exhibits an exceptional understanding of real-life scenario-based project development and accurately identifies all functional requirements. The project is meticulously developed with thorough attention to detail, reflecting a comprehensive understanding

		insufficiently described.			of Object-Oriented Programming project development activities.
Validation	Fails to demonstrate any understanding or implementation of validation forms in their system. There is no attempt to deal with data validation, and validation requirements are completely ignored or incorrectly applied.	Demonstrates limited understanding of validation forms and data validation techniques. While some attempt may be made to implement validation, it is incomplete or poorly executed, leading to inadequate handling of data validation.	Shows a basic understanding of validation forms and data validation techniques. They attempt to implement validation, but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistent handling of data validation.	Effectively demonstrates the use of validation forms and implements data validation techniques. Validation is mostly accurate and comprehensive, ensuring the proper handling of data input and verification in the system.	Exhibits an exceptional understanding and implementation of validation forms and data validation techniques. Validation is meticulously implemented with thorough attention to detail, ensuring robust data validation procedures and contributing to the overall reliability and integrity of the system.
Verification	Fails to demonstrate any attempt to verify the system data or functional requirements. There is no evidence of understanding or implementation of verification processes, and data flow is not considered.	Demonstrates limited understanding of verification processes and data flow in the system. Verification attempts are incomplete or inaccurate, and there is insufficient consideration given to ensuring data integrity and functionality.	Shows a basic understanding of verification processes and attempts to verify system data. However, verification efforts may be inconsistent or lack thoroughness, and there may be gaps in ensuring proper functional requirements and data flow.	Identifies and verifies system data, ensuring proper functional requirements are met. Verification efforts are mostly accurate and thorough, with attention to ensuring data integrity and appropriate data flow within the system.	Exhibits an exceptional understanding of verification processes and meticulously verifies system data. Verification efforts are comprehensive and precise, with a keen focus on ensuring all functional requirements are met and maintaining proper data flow throughout the system.

CO3: Prepare and Explain a real life desktop based application synthesizing several component of C# along with development tools to adhere the given requirements.

Assessment Criteria	Not Attended/ Incorrect (0)	Inadequate (1-2)	Average (3)	Good (4)	Excellent (5)

Evaluation Criteria	**Evaluation Definition**				
Organization of the application	Fails to identify any suitable real time application or requirements for project development activities related to OOP.	Limited understanding about the project scopes and scenarios or identification of functional requirements.	Lacks depth or relevance to OOP project development activities and may contain inaccuracies. Real-life scenarios are mentioned, but the discussion lacks depth or clarity.	Consider and integrate the idea of several core aspects of the project along with relevance to real-life scenarios. Demonstrating a solid understanding of the application presentation.	Generalize and exhibits an exceptional understanding of project preparation according to a to real-life scenarios. Also contains proper and insightful identification of the system which is comprehensive and precise.
Representation and Integration of Database	Fails to identify and present any understanding or implementation of database. Also failed to integrate the data with the project itself.	Limited understanding of the database concepts or their proper way of using in a real time project. While some attempt may be made to implement but it is incomplete or poorly executed, leading to inadequate design.	Lacks depth or relevance to database integration with the application. Shows a basic understanding but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistency. May lack proper normalization.	Integrate the database with the forms properly and implements it with proper validation which is mostly accurate and comprehensive, ensuring the proper handling of data input and verification along with general normalization.	Exhibits an exceptional understanding and implementation of database ensuring attention to detail, and robust data manipulation procedures and contributing to the overall clarity.
Graphical User Interface	Fails to present or prepare GUI based application interfaces. There is no evidence of creating or integrating such things according to their usefulness.	Limited understanding of graphical user interfaces. Lack of design knowledge. Very poor attempt to make such things which are currently obsolete or can't be identified as coherent.	Shows a basic understanding of creating user interfaces. Most of them are interconnected but maybe some of them lack it. However, most of it can be described as user friendly.	Effectively identifies and meet the consider the simplicity. Design related works are mostly accurate and taken proper attention to ensuring a user-friendly coherent system.	Exhibits an exceptional work design following a high standard of simple and elegant work. Several controls and mechanism has been organized in a preferred way according to the coherent usage .

CO4: Perform and solve the proper system functional requirements which combines several aspects of the work to display and verify a real-life based application.

Assessment Criteria	**Not Attended/ Incorrect (0)**	**Inadequate (1-2)**	**Average (3)**	**Good (4)**	**Excellent (5)**

Evaluation Criteria	Evaluation Definition				
Requirement fulfillment	Fails to demonstrate any attempt to perform or solve system functional requirements. There is no evidence of addressing real-life scenario-based projects or identifying functional requirements for Object-Oriented Programming project development activities.	Demonstrates limited understanding of system functional requirements and real-life scenario-based projects. Attempts to perform or solve functional requirements may be incomplete or inaccurately applied, and there is insufficient consideration given to proper identification of requirements.	Shows a basic understanding of system functional requirements and attempts to perform or solve them. However, the execution may lack depth or thoroughness, and there may be gaps in addressing all aspects of real-life scenario-based projects or properly identifying functional requirements.	Effectively demonstrates the performance and solution of system functional requirements for real-life scenario-based projects. Requirements are identified with clarity and relevance, and the execution is mostly accurate and comprehensive, reflecting a solid understanding of Object-Oriented Programming project development activities.	Exhibits an exceptional understanding of system functional requirements and adeptly performs and solves them for real-life scenario-based projects. Requirements are meticulously identified and executed with thorough attention to detail, showcasing a comprehensive mastery of Object-Oriented Programming project development activities.
Validation	Fails to demonstrate any understanding or implementation of validation forms or data validation techniques. There is no evidence of addressing validation requirements or dealing with data validation in their system.	Demonstrates limited understanding of validation forms and data validation techniques. While some attempt may be made to implement validation, it is incomplete or poorly executed, leading to inadequate handling of data validation.	Shows a basic understanding of validation forms and data validation techniques. They attempt to implement validation, but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistent handling of data validation.	Demonstrates the use of validation forms and implements data validation techniques. Validation is mostly accurate and comprehensive, ensuring the proper handling of data input and verification in the system, although there may be minor issues.	Exhibits an exceptional understanding and implementation of validation forms and data validation techniques. Validation is meticulously implemented with thorough attention to detail, ensuring robust data validation procedures and contributing to the overall reliability and integrity of the system.
Verification	Fails to demonstrate any understanding	Demonstrates limited understanding of verification	Shows a basic understanding of verification processes and	Effectively identifies and verifies system data, ensuring	Exhibits an exceptional understanding of verification

	or attempt to verify system data or functional requirements. There is no evidence of addressing verification requirements or considering data flow in their system solution.	processes and data flow in the system. Verification attempts may be incomplete or inaccurately applied, and there is insufficient consideration given to ensuring proper functional requirements and data flow.	attempts to verify system data. However, verification efforts may lack thoroughness or precision, and there may be gaps in ensuring proper functional requirements and data flow within the system.	proper functional requirements are met. Verification efforts are mostly accurate and comprehensive, with attention to ensuring data integrity and appropriate data flow within the system.	processes and meticulously verifies system data. Verification efforts are comprehensive and precise, with a keen focus on ensuring all functional requirements are met and maintaining proper data flow throughout the system.

XIV- Course Requirements

- Students are expected to attend at least 80% of the class.
- Students are expected to participate actively in the class.
- For both terms, there will be at least 2 quizzes based on the theoretical knowledge and conceptual understanding of the topic covered discussed in the classes.
- Submit report based on the given course related problems.
- Submission of assignments and projects should be in due time.

XV – Evaluation & Grading System*

The following grading system will be strictly followed in this class.

MID TERM		FINAL TERM	
Attendance	05%	Attendance	05%
Quiz	30%	Quiz	25%
Lab performance	15%	Lab performance	10%
Midterm	50%	Project, Report and Viva	60%
Total	**100%**	**Total**	**100%**
Grand Total 100% = 40% of Midterm + 60% of Final Term			

Letter	Grade Point	Numerical %
A+	4.00	90-100
A	3.75	85 - < 90
B+	3.50	80 - < 85
B	3.25	75 - < 80
C+	3.00	70 - < 75
C	2.75	65 - < 70
D+	2.50	60 - < 65
D	2.25	50 - < 60

F	0.00	< 50
I		Incomplete
W		Withdrawal
UW		Unofficially Withdrawal

** The evaluation system will be strictly followed as par with the AIUB grading policy.*
** CO attainment will be achieved with 60% of the evaluation marks.*

XVI – Textbook/ References

1. C# 4.0 The Complete Reference; Herbert Schildt; McGraw-Hill Osborne Media; 2010
2. Head First C# by Andrew Stellman
3. The C# Programming Language (Covering C# 4.0); Anders Hejlsberg, Mads Torgersen, Scott Wiltamuth, Peter Golde; Addison-WesleY; 2010
4. MSDN Library; URL: http://msdn.microsoft.com/library
5. C# Language Specification, URL: http://download.microsoft.com/download/0/B/D/0BDA894F- 2CCD-4C2C-B5A7-4EB1171962E5/CSharp%20Language%20Specification.doc
6. Fundamentals of Computer Programming with CSharp – Nakov v2013

XVII - List of Faculties Teaching the Course

FACULTY NAME	SIGNATURE
1. MD. HASIBUL HASAN	
2. TOFAYET SULTAN	
3. AL JUBAIR HOSSAIN	
4. ARJUN KUMAR BOSE ARNOB	
5. AYMED HOSSAIN	
6. DR. MD IFTEKHARUL MOBIN	
7. KAZI SADIA	
8. M. ABRAR FAHAD	
9. MD SADIK TASRIF ANUBHOVE	
10. MD. RAIHAN TALUKDER	
11. MEHEDI SAYED	
12. SANJIDA AFROZ SHIMU	
13. ZAHIDUDDIN AHMED	
14. ZISHAN AHMED ONIK	

XVI – Verification

Prepared by: --------------------------------- **Tofayet Sultan** *Course Convener* Date:.......................................	**Moderated by:** --------------------------------- **Dr. M. Mahmudul Hasan** *Point Of Contact* *OBE Implementation Committee* Date:.......................................	**Checked by:** --------------------------------- **Prof. Dr. Muhammad Firoz Mridha** *Head (Undergraduate Program)* *Department of Computer Science* Date:.......................................

Verified by:	**Certified by:**	**Approved by:**
.... **Dr. Md. Abdullah-Al-Jubair** *Director* *Faculty of Science & Information Technology* Date:...	.... **Prof. Dr. Dip Nandi** *Associate Dean*, *Faculty of Science & Information Technology* Date:...	.... **Mr. Mashiour Rahman** *Dean*, *Faculty of Science & Information Technology* Date:..